

Lecture 9 – Rao–Blackwellization

Last Modified: September 22, 2026

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9.1 Review: conditional expectation

Let (X, Y) be a continuous random vector. Recall that its joint density is $f_{X,Y}(x, y)$, its marginal density is

$$f_X(x) = \int f_{X,Y}(x, y) \, dy,$$

and the conditional density of X given $Y = y$ is

$$f_{X|Y}(x | y) = \frac{f_{X,Y}(x, y)}{f_Y(y)}.$$

The conditional expectation of $g(X)$ given $Y = y$ is

$$\mathbb{E}[g(X) | Y = y] = \int g(x) f_{X|Y}(x | y) \, dx.$$

Example 1 (A conditional expectation). Suppose $X_1, X_2 \stackrel{\text{iid}}{\sim} \text{Exp}(1)$ and let $Y = X_1 + X_2$. Our goal is to derive the conditional density of X_1 given $Y = y$. We know that $Y \sim \text{Gamma}(2, 1)$, so

$$f_Y(y) = ye^{-y} \mathbb{1}_{\{y \geq 0\}}.$$

We first find the conditional density of Y given $X_1 = x$. Once $X_1 = x$ is fixed, $Y = x + X_2 = x + \text{Exp}(1)$. Thus Y is in a location family, and we worked out before how to get its density:

$$f_{Y|X_1}(y | x) = e^{-(y-x)} \mathbb{1}_{\{y-x \geq 0\}} = e^{-(y-x)} \mathbb{1}_{\{y \geq x\}}.$$

We can now obtain the joint density by multiplying the conditional density by the marginal density of X_1 :

$$\begin{aligned} f_{X_1,Y}(x, y) &= f_{Y|X_1}(y | x) f_{X_1}(x) \\ &= e^{-(y-x)} e^{-x} \mathbb{1}_{\{x \geq 0, y \geq x\}} \\ &= e^{-y} \mathbb{1}_{\{0 \leq x \leq y\}}. \end{aligned}$$

It follows that

$$f_{X_1|Y}(x | y) = \frac{f_{X_1,Y}(x, y)}{f_Y(y)} = \frac{e^{-y}}{ye^{-y}} \mathbb{1}_{\{0 \leq x \leq y\}} = \frac{1}{y} \mathbb{1}_{\{0 \leq x \leq y\}}.$$

Thus,

$$X_1 | Y = y \sim \text{Unif}(0, y)$$

and

$$\mathbb{E}[X_1 \mid Y = y] = \frac{1}{2}y.$$

Note that the conditional expectation given $Y = y$ is a function of y . Evaluating this function at the random variable Y gives a random variable:

$$\mathbb{E}[X_1 \mid Y] = \frac{1}{2}Y.$$

More generally, take any $U = U(X_1, X_2)$. Then

$$\mathbb{E}[U(X_1, X_2) \mid Y = y] = h(y),$$

for some function h . Hence

$$\mathbb{E}[U(X_1, X_2) \mid Y] = h(Y)$$

is a random variable and a function of $Y = X_1 + X_2$.

Theorem 1 (Tower property and total variance). Let (X, Y) be a random vector, with X having finite second moment. Then:

- (i) The **tower property** (or law of total expectation) is

$$\mathbb{E}[X] = \mathbb{E}[\mathbb{E}[X \mid Y]].$$

- (ii) The **law of total variance** is

$$\text{Var}(X) = \text{Var}(\mathbb{E}[X \mid Y]) + \mathbb{E}[\text{Var}(X \mid Y)].$$

The tower property says that averaging in two stages gives the same overall mean. For continuous random variables, it follows directly from the conditional density:

$$\begin{aligned} \mathbb{E}[\mathbb{E}[X \mid Y]] &= \int \left(\int x f_{X|Y}(x \mid y) dx \right) f_Y(y) dy \\ &= \int \int x f_{X,Y}(x, y) dx dy \\ &= \int x f_X(x) dx = \mathbb{E}[X]. \end{aligned}$$

The law of total variance splits the variation in X into

- variation of the conditional averages across different values of Y , and
- variation of X among outcomes with the same value of Y .

In particular,

$$\text{Var}(\mathbb{E}[X \mid Y]) = \text{Var}(X) - \underbrace{\mathbb{E}[\text{Var}(X \mid Y)]}_{\text{variation averaged out}} \leq \text{Var}(X).$$

Proof of the total-variance identity. Write $m(Y) = \mathbb{E}[X \mid Y]$. Conditional on Y , add and subtract $m(Y)$:

$$\begin{aligned} \mathbb{E}[X^2 \mid Y] &= \mathbb{E}[(X - m(Y) + m(Y))^2 \mid Y] \\ &= \mathbb{E}[(X - m(Y))^2 \mid Y] + m(Y)^2 \\ &= \text{Var}(X \mid Y) + \mathbb{E}[X \mid Y]^2. \end{aligned}$$

Taking expectations gives

$$\mathbb{E}[X^2] = \mathbb{E}[\text{Var}(X \mid Y)] + \mathbb{E}[\mathbb{E}[X \mid Y]^2].$$

Subtracting $\mathbb{E}[X]^2 = \mathbb{E}[\mathbb{E}[X \mid Y]]^2$ from both sides gives the result. $\square$

9.2 Motivation for Rao–Blackwellization

In Lecture 8, we defined a uniformly minimum-variance unbiased estimator (UMVUE). If U is an unbiased estimator of $g(\theta)$, how can we systematically construct another unbiased estimator with smaller variance?

The two key observations are:

1. **Averaging reduces variance.** For any statistic S , the law of total variance gives

$$\text{Var}_\theta(\mathbb{E}_\theta[U \mid S]) \leq \text{Var}_\theta(U).$$

The law of total expectation also shows that this averaging preserves the mean.

2. **The averaged estimator must not depend on the unknown parameter.** For an arbitrary statistic S , the conditional expectation may have the form $h_\theta(S)$ and therefore may not be computable from the data alone. If instead we condition on a sufficient statistic T , then the conditional distribution of $\underline{X}$ given T does not depend on θ . Consequently, for every statistic $U = U(\underline{X})$,

$$\mathbb{E}_\theta[U \mid T] = h(T)$$

for a function h that does not depend on θ .

This is the crux of Rao–Blackwellization: conditional averaging gives the variance reduction, while sufficiency ensures that the result is a genuine statistic.

The method is especially useful because it may be easy to recognize some estimator U satisfying

$$\mathbb{E}_\theta[U] = g(\theta),$$

even when it is hard to directly solve for an unbiased estimator of the form $h(T)$. The strategy is

find any convenient unbiased U	$\longrightarrow$	$\mathbb{E}_\theta[U \mid T].$
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The initial estimator is a convenient starting point; it need not be the estimator we ultimately use.

9.3 The Rao–Blackwell theorem

Theorem 2 (Rao–Blackwell). Let $U = U(\underline{X})$ have finite second moment, and let $T = T(\underline{X})$ be sufficient for θ . Define

$$U^* = \mathbb{E}_\theta[U \mid T].$$

Then:

- (i) U^* is a statistic: it is a function of T that does not depend on the unknown θ ;
- (ii) $\mathbb{E}_\theta[U^*] = \mathbb{E}_\theta[U]$ for every θ ;
- (iii) $\text{Var}_\theta(U^*) \leq \text{Var}_\theta(U)$ for every θ ;
- (iv) for any scalar target $g(\theta)$, under squared-error loss, $\text{MSE}_{U^*}(\theta) \leq \text{MSE}_U(\theta)$ for every θ .

Equality in the variance comparison at a particular θ holds if and only if $U = U^*$ with probability one under that θ .

Proof. Because T is sufficient, the conditional distribution of $\underline{X}$ given T does not depend on θ . Since U is a function of $\underline{X}$, the conditional distribution of U given T also does not depend on θ . Therefore $U^* = \mathbb{E}_\theta[U \mid T]$ can be calculated as a function of T without knowing θ . The law of total expectation gives

$$\mathbb{E}_\theta[U^*] = \mathbb{E}_\theta[\mathbb{E}_\theta[U \mid T]] = \mathbb{E}_\theta[U].$$

Thus U and U^* have the same bias. The law of total variance gives

$$\text{Var}_\theta(U) = \text{Var}_\theta(U^*) + \mathbb{E}_\theta[\text{Var}_\theta(U \mid T)] \geq \text{Var}_\theta(U^*).$$

Combining the same bias with the smaller variance proves the MSE statement. Equality holds exactly when $\text{Var}_\theta(U \mid T) = 0$ with probability one, which is equivalent to $U = \mathbb{E}_\theta[U \mid T] = U^*$ with probability one. $\square$

If U is unbiased for $g(\theta)$, then part (ii) shows that U^* is also unbiased for $g(\theta)$. Since MSE equals variance for unbiased estimators, Rao–Blackwellization produces an estimator that is at least as good at every θ within the class considered in Lecture 8.

9.4 A Poisson example

Example 2 (Replace indicators by a conditional probability). Suppose

$$X_1, \dots, X_n \stackrel{\text{iid}}{\sim} \text{Poisson}(\theta), \quad n \geq 2,$$

and we want to estimate $g(\theta) = \mathbb{P}_\theta(X_1 = 0) = e^{-\theta}$. Let

$$U = \frac{1}{n} \sum_{i=1}^n \mathbb{1}_{\{X_i=0\}}$$

is an unbiased estimator. Let $T = \sum_{i=1}^n X_i$, which is sufficient for θ . To derive the conditional distribution of X_1 given $T = t$, write

$$T = X_1 + \sum_{i=2}^n X_i,$$

where $X_1 \sim \text{Poisson}(\theta)$ is independent of $\sum_{i=2}^n X_i \sim \text{Poisson}((n-1)\theta)$. For $k = 0, \dots, t$,

$$\begin{aligned} \mathbb{P}_\theta(X_1 = k \mid T = t) &= \frac{\mathbb{P}_\theta(X_1 = k, \sum_{i=2}^n X_i = t - k)}{\mathbb{P}_\theta(T = t)} \\ &= \frac{\frac{e^{-\theta} \theta^k}{k!} \frac{e^{-(n-1)\theta} ((n-1)\theta)^{t-k}}{(t-k)!}}{\frac{e^{-n\theta} (n\theta)^t}{t!}} \\ &= \binom{t}{k} \left(\frac{1}{n}\right)^k \left(1 - \frac{1}{n}\right)^{t-k}. \end{aligned}$$

Thus

$$X_1 \mid T = t \sim \text{Bin}\left(t, \frac{1}{n}\right),$$

and in particular

$$\mathbb{P}_\theta(X_1 = 0 \mid T = t) = \left(1 - \frac{1}{n}\right)^t.$$

By exchangeability, the same conditional probability holds for every X_i . Therefore,

$$\begin{aligned} U^* &= \mathbb{E}_\theta \left[\frac{1}{n} \sum_{i=1}^n \mathbb{1}_{\{X_i=0\}} \mid T \right] \\ &= \frac{1}{n} \sum_{i=1}^n \mathbb{P}_\theta(X_i = 0 \mid T) \\ &= \mathbb{P}_\theta(X_1 = 0 \mid T) \\ &= \left(1 - \frac{1}{n}\right)^T. \end{aligned}$$

The original estimator averages the observed zero indicators. Its Rao–Blackwellization replaces that average by the conditional probability of a zero given the total count.

How does this compare with the plug-in estimator? The MLE of θ is

$$\hat{\theta}_{\text{MLE}} = \bar{X}_n = \frac{T}{n},$$

so the plug-in estimator is

$$g(\hat{\theta}_{\text{MLE}}) = e^{-\bar{X}_n} = e^{-T/n} = (e^{-1/n})^T.$$

Both estimators are functions of the sufficient statistic T , and they are close when n is large, but they are not equal:

$$U^* = \left(1 - \frac{1}{n}\right)^T \quad \text{whereas} \quad g(\hat{\theta}_{\text{MLE}}) = (e^{-1/n})^T.$$

The Rao–Blackwellized estimator is exactly unbiased, while the plug-in MLE is upward biased. The approximation $\log(1 - 1/n) \approx -1/n$ explains why they are close for large n .

9.5 Conditioning on a minimal sufficient statistic

Suppose S is minimal sufficient and T is any sufficient statistic. By minimal sufficiency, S is a function of T . Intuitively, $\mathbb{E}_\theta[U \mid T]$ assigns an average value of U to each possible value of T . Because S is coarser, several values of T may correspond to the same value of S . Conditioning again on S averages those finer T -group means together. Merging groups and averaging their means removes between-group variation, so the resulting conditional expectation cannot be more variable. Specifically, one can show that

$$\text{Var}_\theta(\mathbb{E}_\theta[U \mid S]) \leq \text{Var}_\theta(\mathbb{E}_\theta[U \mid T]).$$

Thus you'll decrease the variance more by conditioning on the coarsest possible sufficient statistic, i.e. on the minimal sufficient statistic.